MERGE: Multimodal Emotion-Driven Rhythmic Lyric Generation Engine

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Abstract

Creative language generation has gained increasing attention in natural language processing, particularly for tasks such as poetry and song lyric generation. However, most existing systems focus either on structural constraints such as rhyme and meter or on semantic control mechanisms such as sentiment, topic, or style. The integration of multimodal emotional signals with rhythm-aware lyric generation remains relatively underexplored.

In this work, we propose **MERGE** (Multimodal Emotion-Driven Rhythmic Lyric Generation Engine), a framework for generating rhythm-aware song lyrics conditioned on emotional states inferred from facial expressions. The system first detects the user's emotional state using a facial emotion recognition model and then conditions a transformer-based language model on the predicted emotion to guide lyric generation. To maintain lyrical structure, rhythm-aware constrained decoding is applied to enforce syllable counts and rhyme patterns during generation.

We further introduce **PHREM**, a composite evaluation metric designed for lyric generation that measures rhythmic consistency, emotional alignment, lexical diversity, and semantic coherence. MERGE provides a framework for exploring emotion-conditioned lyric generation while preserving structural properties of song lyrics.

1 Introduction

The motivation of this project is to develop an emotion-aware system capable of generating creative musical content based on a user's emotional state inferred from sensory inputs. Direct music synthesis conditioned on physiological or multimodal sensor data poses significant challenges due to limited dataset availability, the high dimensionality of audio generation, and the complex alignment between emotional intent and musical structure (Cao et al., 2022). To preserve the objective of emotion-driven creative generation while remaining within the methodological scope of natural language processing (NLP), this work reformulates the problem as emotion-conditioned lyric generation.

Facial emotion recognition provides a practical and widely studied modality for sensing emotional states (Goodfellow et al., 2013; Li et al., 2017). At the same time, song lyrics serve as a primary textual medium through which emotional expression is conveyed in music. These properties make lyrics a suitable domain for studying emotion-controlled and structurally constrained language generation. Prior research in NLP has explored emotion-conditioned text generation as well as structural constraints such as rhyme schemes and syllable patterns in poetry and lyric generation. However, these directions have largely been investigated independently.

A key challenge in emotion-conditioned lyric generation lies in the limited availability of large-scale lyric datasets with explicit emotion annotations. While curated datasets such as the Spotify Million Playlist Dataset (Devdope, 2023) provide approximately 500k lyrics annotated with emotional labels, their scale remains limited for training modern neural language models. To address this limitation, this work leverages large-scale lyric corpora obtained from the Genius lyric dataset (Lim and Benson, 2021), which provides substantially larger collections of song lyrics. Since explicit emotion annotations are not available for these datasets, we construct emotion labels using a scalable annotation pipeline that aggregates predictions from a GoEmotions-based classifier and maps fine-grained emotional categories into the six facial emotion recognition (FER) classes through normalization and weighted line-level aggregation. This procedure enables the creation of large-scale emotion-labeled lyric datasets suitable for training emotion-

conditioned language models.

While prior work has examined controlled text generation and rhythm-aware lyric generation separately, the integration of facial emotion signals as control inputs for rhythm-aware lyric generation remains relatively underexplored, particularly in English-language songwriting. Bridging this gap requires a framework that can incorporate visual emotional cues into the generation process while maintaining the structural characteristics that define song lyrics.

In this work, we propose **MERGE (Multimodal Emotion-Driven Rhythmic Lyric Generation Engine)**, a multimodal framework that connects facial emotion recognition with rhythm-aware lyric generation. MERGE uses facial emotion recognition models to infer emotional states from visual inputs and employs these signals to condition a transformer-based language model trained on large-scale lyric corpora. To ensure structural coherence, the generation process incorporates rhythm-aware decoding constraints based on syllable counts and rhyme schemes.

This work investigates two central research questions: (1) whether facial emotion signals can effectively serve as control inputs for lyric generation, and (2) whether rhythmic constraints such as syllable count and rhyme can be enforced during decoding in emotion-conditioned lyric generation without degrading emotional alignment or linguistic fluency. The contributions of this work are fourfold: (1) MERGE, a multimodal framework that connects facial emotion recognition with lyric generation, (2) a scalable lyric emotion annotation pipeline that maps fine-grained GoEmotions predictions to facial emotion recognition (FER) categories using normalization and weighted aggregation, enabling large-scale emotion labeling of lyric corpora, (3) a rhythm-aware constrained decoding strategy for maintaining lyrical structure during generation, and (4) REDS, a composite evaluation metric designed for assessing emotional alignment and structural quality in generated lyrics.

The source code and implementation details for this project are available at the following link.[1]

2 Related Work

Facial Emotion Recognition. Facial emotion recognition has achieved strong performance using deep convolutional neural networks, with architectures such as ResNet and MobileNet commonly evaluated on benchmark datasets including FER-2013 and RAF-DB (Goodfellow et al., 2013; Li et al., 2017). The FER-2013 dataset, introduced as part of a large-scale representation learning challenge, contains over 35,000 grayscale facial images annotated with basic emotion labels such as angry, disgust, fear, happy, sad, surprise, and neutral (Goodfellow et al., 2013). Due to its accessibility and scale, FER-2013 remains one of the most widely used benchmarks for training and evaluating facial emotion recognition systems. More recent research has explored transformer-based architectures and multimodal fusion networks for emotion recognition, improving the ability of models to capture contextual emotional signals from visual data (Zhang et al., 2023). These advances make facial emotion recognition sufficiently reliable to serve as an emotion sensing component in downstream generative systems.

Controlled Text Generation. In natural language processing, controlled text generation has been widely studied as a mechanism for steering language models toward desired attributes such as sentiment, topic, or style. Early approaches relied on conditional fine-tuning strategies and control tokens for large language models (Radford et al., 2019). More recent work has explored attribute-guided generation techniques that allow large language models to produce text aligned with specific semantic properties, including emotional tone and stylistic constraints (Hu et al., 2023). These approaches demonstrate that external control signals can effectively influence the behavior of generative language models.

Lyric and Poetry Generation. Research on poetry and lyric generation has extensively explored methods for enforcing structural constraints such as rhyme schemes, meter, and syllable counts. Early approaches relied on rule-based techniques and phonetic resources such as the CMU Pronouncing Dictionary to analyze syllable patterns and rhyming structures (Ghazvininejad et al., 2016; Gonçalo Oliveira, 2024). More recent neural approaches incorporate such constraints during generation to produce structurally coherent poetic text. However, most existing systems primarily focus on structural control and do not explicitly i nvestigate how these rhythmic constraints interact with emotion-conditioned generation, particularly when emotional signals originate from exter-

[1] https://github.com/IIITH-2025-27/NLP_Project

nal modalities such as facial expressions. Neural approaches later introduced sequence-to-sequence and transformer-based architectures capable of generating structured poetic text. More recent lyric generation systems incorporate musical or prosodic signals to guide the generation process. For example, SongRewriter (Sun et al., 2023) employs a randomized multi-level masking strategy to enable both lyric generation and editing while incorporating keyword prompts and vowel modeling tasks to encourage flexible rhyme schemes. XAI-Lyricist (Liang et al., 2024) conditions lyric generation on musical prosody signals to improve singability while also providing interpretable explanations for rhythmic alignment.

Collaborative and Multimodal Generation Systems. Another emerging direction explores collaborative or multi-agent architectures for lyric generation. Liu et al. (Liu and Liu, 2024) propose a multi-agent system in which different agents handle specialized tasks such as rhyme suggestion, syllable control, and lyric–melody alignment. Similarly, Ram et al. (Ram et al., 2021) employ multitask transfer learning with the T5 transformer model to model lyrical constraints including beat alignment and target word endings. More broadly, recent multimodal generative models such as MusicLM (Huang et al., 2023) demonstrate the potential of combining semantic signals with generative models for music-related tasks, although these systems primarily focus on audio generation rather than lyric generation.

Despite these advances, most existing systems either focus on structural lyric generation or rely on musical signals such as melody or prosody as conditioning inputs. The integration of visual emotion signals—such as facial expressions—with rhythm-aware lyric generation remains relatively underexplored, particularly in English-language songwriting. MERGE addresses this gap by providing an end-to-end framework that connects facial emotion recognition with constrained lyric generation.

3 Dataset Details

3.1 Lyric Corpus

Lyrics are sourced from the **Genius** corpus (Lim and Benson, 2021), which contains songs spanning many languages. After filtering for English-language content, approximately **3.2 million** songs remain. A cleaned version of this corpus is publicly available on Kaggle.[2]

3.2 Facial Expression Datasets

The FER component is trained on a unified corpus comprising three publicly available datasets:

- **FER-2013** (Goodfellow et al., 2013): 35,887 grayscale 48×48 images, seven original classes (disgust excluded, yielding six classes).
- **RAF-DB** (Li et al., 2017): ∼15,339 real-world images with basic and compound emotion labels (disgust excluded), It was found on kaggle.[3]
- **AffectNet** (Mollahosseini et al., 2017): ∼450,000 images annotated with eight emotions. But out of them only ∼30,000 was used for training (disgust excluded), which were found on kaggle.[4]

A unified six-class mapping $\mathcal{E} = \{angry, fear, happy, sad, surprise, neutral\}$ is defined across all three sources. The disgust and contempt categories are excluded because they appear rarely in song lyrics and their linguistic expression substantially overlaps with anger and sadness. Additionally, FER-2013 and other datasets contain very few disgust, contempt, and surprise samples, resulting in a severe class imbalance.

3.3 Dataset Construction and Emotion Mapping

Large-scale lyric datasets typically lack explicit emotion annotations, which makes it difficult to train emotion-conditioned language models. To address this limitation, we construct an emotion-labeled lyric dataset using an automatic annotation pipeline as shown in Figure 1.

Lyrics are collected from large public lyric corpora, specifically the Genius dataset (Lim and Benson, 2021), which contains lyrics in many languages out of which approx 3.2 million belongs to English language. Emotional signals are inferred using a pretrained emotion classifier based on the

[2] https://www.kaggle.com/datasets/ankit477/genius-dataset-cleaned
[3] https://www.kaggle.com/datasets/shuvoalok/raf-db-dataset
[4] https://www.kaggle.com/datasets/mstjebashazida/affectnet

RoBERTa architecture(Lowe, 2023), which is fine-tuned on the GoEmotions dataset (Demszky et al., 2020). The GoEmotions label space contains 28 categories, consisting of 27 fine-grained emotion labels and one neutral category. For each input lyric line, the classifier outputs a probability distribution over these 28 labels, which serves as the basis for constructing the emotion representations used in the subsequent stages of the pipeline.

To ensure compatibility between textual emotions and facial emotion recognition outputs, the fine-grained emotion space is mapped into a smaller set of basic emotions, inspired by the circumplex model of affect and theories of basic emotions (Russell, 1980; Ekman, 1992). The emotion space is reduced to the following set

$$\mathcal{E} = \{\text{angry}, \text{fear}, \text{happy}, \text{sad}, \text{surprise}, \text{neutral}\}.$$

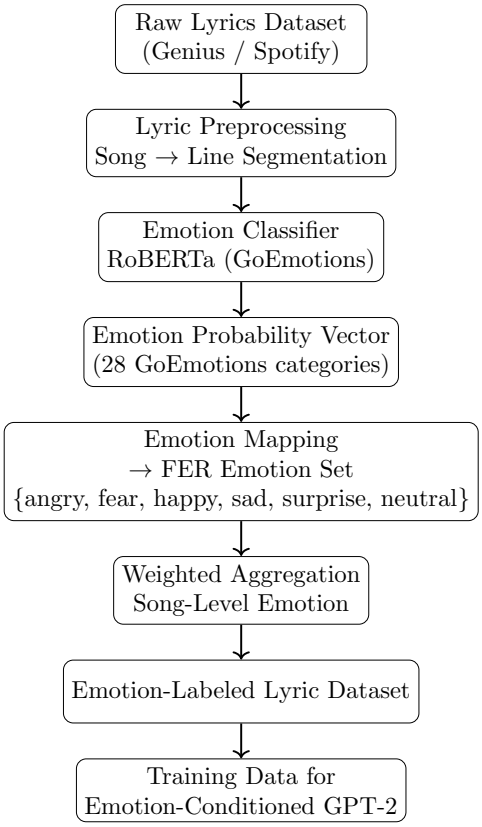

Figure 1: Pipeline for constructing the emotion-labeled lyric dataset.

The disgust category is excluded for two reasons. First, explicit expressions of disgust rarely occur in song lyrics and are often linguistically expressed as sadness or disappointment. Second, the FER-2013 facial emotion dataset contains very few samples labeled as disgust, resulting in severe class imbalance during model training. To maintain a consistent emotion space across modalities, disgust-related signals are therefore merged into the sadness category. The created dataset is being hosted on kaggle(Kumar et al., 2025).

3.4 Lyric Dataset and Emotion Labeling

Each song is segmented into individual lyric lines. Let a song contain n lyric lines, where the index of a line is denoted by

$$i \in \{1, 2, \ldots, n\}.$$

For each lyric line i, the GoEmotions classifier produces a probability vector over the 28 emotion labels:

$$\mathbf{p}_i = \left[p_i^{(1)}, p_i^{(2)}, \ldots, p_i^{(28)}\right] \in \mathbb{R}^{28},$$

where $p_i^{(e)}$ denotes the probability assigned to label e and satisfies

$$0 \le p_i^{(e)} \le 1.$$

To obtain emotion representations compatible with facial emotion recognition categories, the fine-grained emotion labels are grouped into six emotion sets:

$$\begin{aligned}
A &= \{\text{anger}, \text{annoyance}, \text{disapproval}\},\\
F &= \{\text{fear}, \text{nervousness}\},\\
H &= \{\text{admiration}, \text{amusement}, \text{approval}, \text{caring},\\
&\quad \text{desire}, \text{excitement}, \text{gratitude}, \text{joy}, \text{optimism},\\
&\quad \text{love}, \text{pride}, \text{relief}\},\\
S &= \{\text{sadness}, \text{grief}, \text{disappointment}, \text{remorse},\\
&\quad \text{disgust}\},\\
U &= \{\text{surprise}, \text{realization}\},\\
N &= \{\text{neutral}, \text{curiosity}, \text{confusion}\}.
\end{aligned}$$

Using these groups, the emotion score for each category is computed by summing probabilities belonging to the same group. For example, the anger score for line i is defined as

$$q_i^{\text{angry}} = \sum_{e \in A} p_i^{(e)}.$$

Applying the same procedure to all emotion groups produces a six-dimensional emotion vector for lyric line i:

$$\mathbf{q}_i = \left[q_i^{\text{angry}}, q_i^{\text{fear}}, q_i^{\text{happy}}, q_i^{\text{sad}}, q_i^{\text{surprise}}, q_i^{\text{neutral}}\right].$$

Because multiple probabilities are aggregated, the resulting vector is normalized to obtain a valid probability distribution:

$$\mathbf{q}_i = \frac{\mathbf{q}_i}{\sum_{k=1}^{6} q_i^{(k)}}.$$

Lyrics often contain narrative lines with little emotional content. To reduce their influence, each line is assigned an emotional weight defined as

$$w_i = 1 - q_i^{\text{neutral}},$$

where w_i represents the emotional intensity of lyric line i.

The final emotion representation of a song is computed by aggregating the line-level emotion vectors using weighted averaging:

$$\mathbf{E}_{\text{song}} = \frac{\sum_{i=1}^{n} w_i \mathbf{q}_i}{\sum_{i=1}^{n} w_i}.$$

The dominant emotion of the song is defined as

$$e^* = \arg\max_k \mathbf{E}_{\text{song}}^{(k)},$$

where $\mathbf{E}_{\text{song}}^{(k)}$ denotes the score of emotion category k.

4 Methodology

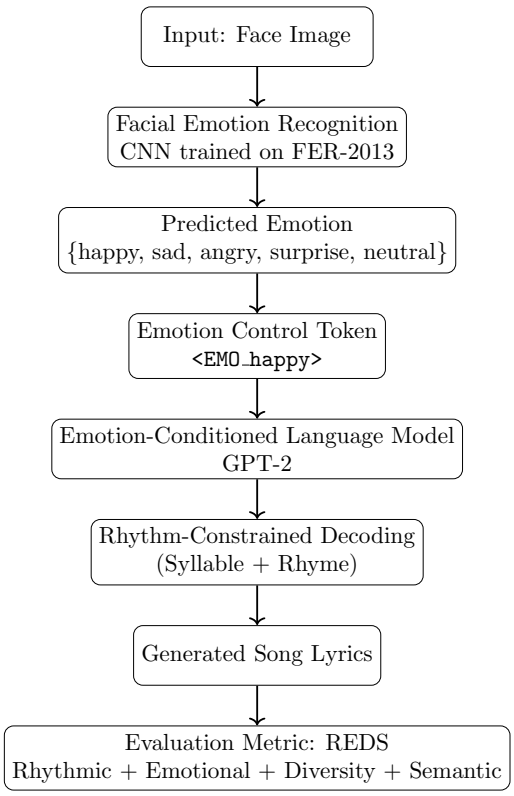

Figure 2: Overview of the MERGE framework. A face image is processed by the FER ensemble to produce an emotion control token, which is prepended to the lyric sequence for GPT-2 fine-tuning. Rhythm-constrained decoding enforces syllable counts and rhyme schemes at inference.

This work proposes an end-to-end multimodal framework that connects facial emotion recognition with emotion-conditioned and rhythm-aware lyric generation. The system (as shown in Figure 2) first detects emotional states from facial inputs and then uses these signals to guide lyric generation while enforcing linguistic and rhythmic constraints such as rhyme patterns and syllable structure.

4.1 Facial Emotion Conditioning

The MERGE framework uses facial expressions as the emotional input signal for lyric generation. Given a facial image, a facial emotion recognition model predicts the emotional state of the user (as shown in Figure 3), which is then used to condition the lyric generation process.

The facial emotion recognition model is trained on the FER-2013 dataset (Goodfellow et al., 2013), which contains 35,887 grayscale facial images labeled with basic emotion categories. Due to the severe class imbalance of the disgust category in FER-2013, the task is reformulated as a 6-class classification problem. The final emotion set used in this work is angry, fear, happy, sad, surprise, neutral.

Let I denote the input facial image and let f_θ represent the facial emotion recognition model parameterized by θ. The model produces a vector of logits

$$\mathbf{z} = f_\theta(I)$$

where $\mathbf{z} \in \mathbb{R}^6$ represents the unnormalized scores corresponding to the six emotion categories.

The probability distribution over emotions is computed using the softmax function

$$P(e_k \mid I) = \frac{\exp(z_k)}{\sum_{j=1}^{6} \exp(z_j)}$$

where $e_k \in \mathcal{E}$ denotes the k-th emotion category and z_k represents the corresponding logit value.

The predicted facial emotion is obtained by selecting the class with the highest probability

$$e^* = \arg\max_{e_k \in \mathcal{E}} P(e_k \mid I)$$

The predicted emotion e^* is then used as the conditioning signal for the lyric generation model. During inference, this emotion is converted into a control token (for example `<sad>`) which is prepended to the input sequence of the language model. This allows the model to generate lyrics

that are aligned with the emotional state inferred from the user's facial expression.

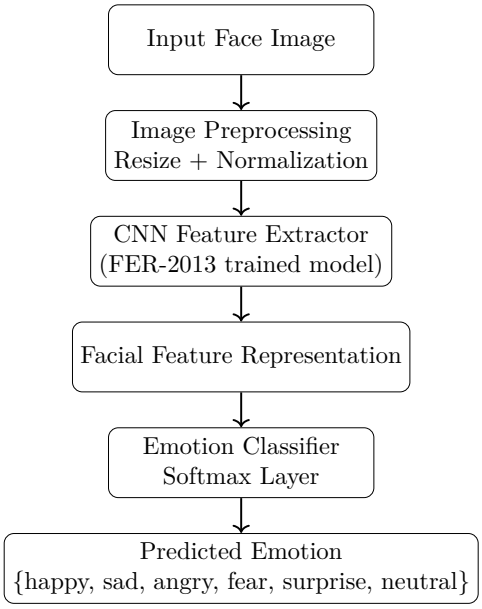

Figure 3: Pipeline of the facial emotion recognition module.

4.2 Emotion-Conditioned Language Model

Lyric generation is formulated as a controlled text generation task in which emotional context influences the semantic content of generated lyrics. We fine-tune the GPT-2 language model (Radford et al., 2019) on the emotion-labeled lyric dataset constructed in the previous subsection.

Emotion conditioning is implemented using explicit control tokens representing the target emotion category. During training, an emotion token (for example `<EMO_sad>`) is prepended to each lyric sequence so that the model can learn associations between emotional states and linguistic patterns.

Let

$$x = (x_1, x_2, \ldots, x_T)$$

denote a lyric token sequence of length T, and let e represent the conditioning emotion label. The model learns the conditional probability

$$P(x \mid e) = \prod_{t=1}^{T} P(x_t \mid x_{<t}, e),$$

where x_t denotes the token generated at step t and $x_{<t}$ represents the previously generated tokens.

The model is trained using the standard cross-entropy objective for next-token prediction.

4.3 Rhythm-Constrained Decoding

To enforce rhythmic and rhyming structure during lyric generation, constrained decoding is applied. The CMU Pronouncing Dictionary provides phonetic representations that allow syllable counting and rhyme detection (CMU, 2014).

Each generated lyric line must satisfy predefined structural constraints such as a target syllable count and a specified rhyme scheme (for example AABB or ABAB). During decoding, candidate tokens that violate these constraints are filtered from the vocabulary.

When a rhyming token is required at the end of a line, the candidate token set is restricted to words whose phonetic representations match the desired rhyme pattern. Hard constraints enforce structural consistency within each line, while soft back-off strategies allow constraint relaxation or line regeneration if decoding becomes infeasible.

By integrating these constraints directly into the decoding process rather than applying them as post-processing steps, the system generates lyrics that maintain both rhythmic structure and semantic coherence.

5 Evaluation Metric: PHREM

Existing metrics (BLEU, ROUGE) measure lexical overlap but fail to capture lyrical essence: phonetic flow, emotional continuity, rhythmic structure, hook catchiness, and semantic coherence. We introduce **PHREM** (Phonetic Rhythm Emotion Metric), a composite evaluation framework designed specifically for emotion-conditioned lyric quality assessment.

PHREM integrates seven complementary dimensions, each capturing distinct aspects of lyric quality. The metric is computed as a weighted sum:

$$\text{PHREM} = \sum_{i=1}^{7} \alpha_i c_i \tag{1}$$

where weights are: $\alpha_{\text{ppfs}} = 0.12$, $\alpha_{\text{rsfs}} = 0.13$, $\alpha_{\text{eas}} = 0.22$, $\alpha_{\text{hqcs}} = 0.13$, $\alpha_{\text{rcs}} = 0.10$, $\alpha_{\text{mcs}} = 0.20$, $\alpha_{\text{dps}} = 0.10$. All components are normalized to $[0, 1]$, ensuring PHREM $\in [0, 1]$.

PPFS: Phonetic Pattern Flow Score. PPFS evaluates the smoothness and naturalness of lyric delivery at the phoneme level. High-quality lyrics exhibit smooth transitions between phonemes without jarring vowel-consonant shifts or stress disruptions. For adjacent phonemes p_i and p_{i+1}:

$$c(p_i, p_{i+1}) = 0.6 \cdot \mathbb{1}[\mathrm{v}(p_i) \neq \mathrm{v}(p_{i+1})] + 0.4 \cdot |s_i - s_{i+1}| \tag{2}$$

where $\mathbb{1}[\cdot]$ indicates vowel-consonant transition (penalty 0.6), and $|s_i - s_{i+1}|$ captures stress mismatch (weight 0.4). The final score is PPFS $= \exp(-\bar{c})$, where $\bar{c}$ is average transition cost. This ensures smooth phonetic flow scores near 1.0 for natural-sounding lyrics.

RSFS: Rhythmic Structure Flow Score. RSFS measures rhythmic consistency by evaluating syllable uniformity and stress pattern regularity across lyric lines. Songs with consistent syllable counts and predictable stress patterns are easier to sing and more memorable. Syllable consistency relative to median line length:

$$R_s = 1 - \frac{1}{n} \sum_{i=1}^{n} \frac{|s_i - \mathrm{med}(s)|}{\mathrm{med}(s)} \tag{3}$$

Stress consistency (R_{stress}) compares consecutive line stress sequences. The final score combines both: RSFS $= 0.7R_s + 0.3R_{\text{stress}}$, prioritizing syllable uniformity while acknowledging stress patterns.

EAS: Emotion Arc Score. EAS measures emotional continuity and coherence across the lyric sequence. For each line i, the GoEmotions classifier (mapped to 6 FER classes) produces normalized emotion vector $\mathbf{q}_i \in [0,1]^6$. Transitions between consecutive lines use Jensen-Shannon divergence:

$$d_i = \frac{1}{2} \left[\mathrm{KL}(\mathbf{q}_i \| \mathbf{m}) + \mathrm{KL}(\mathbf{q}_{i+1} \| \mathbf{m}) \right] \tag{4}$$

where $\mathbf{m} = \frac{1}{2}(\mathbf{q}_i + \mathbf{q}_{i+1})$. Each transition is weighted by emotional intensity $w_i = 1 - q_i^{\text{neutral}}$, downweighting neutral lines. Final score: EAS $= \exp(-\bar{d}_w)$.

HQCS: Hook Quality and Catchiness Score. HQCS identifies and rewards repeated n-gram patterns (hooks) that enhance memorability and catchiness. Hooks are linguistic patterns listeners recognize and remember (e.g., recurring refrains). We extract all 3-grams and count those appearing multiple times. HQCS combines three metrics:

$$\mathrm{HQCS} = 0.4 \cdot \mathrm{sal} + 0.3 \cdot \mathrm{cov} + 0.3 \cdot \mathrm{flow} \tag{5}$$

where salience = max hook frequency / total n-grams, coverage = sum hook frequencies / total n-grams, and flow = mean(PPFS, RSFS). A diversity penalty balances hook repetition against monotony.

RCS: Rhyme Consistency Score. RCS evaluates rhyme quality by comparing phonetic endings of consecutive line-final words. Rhyme quality is determined from the last stressed vowel onward using CMU Pronouncing Dictionary. Similarity between consecutive rhyme patterns is computed by matching phonemes backward from the end. RCS is the mean similarity across all consecutive line pairs.

MCS: Motif Consistency Score. MCS measures semantic coherence and thematic consistency using pre-trained sentence embeddings. High-quality lyrics maintain topical coherence across lines. MCS balances local and global consistency:

$$\mathrm{MCS} = 0.7 \cdot L + 0.3 \cdot G \tag{6}$$

where L = mean cosine similarity of consecutive lines (local coherence), and G = mean similarity of sampled non-adjacent pairs (global thematic consistency).

DPS: Degeneracy Penalty Score. DPS penalizes excessive lexical repetition, indicating degenerate output lacking diversity. The penalty is:

$$\mathrm{DPS} = \exp(-3.0 \cdot r) \tag{7}$$

where $r = \frac{\text{max word frequency}}{\text{total words}}$. Low repetition yields DPS ≈ 1.0; high repetition decays exponentially. This rewards lexical diversity while allowing natural reuse.

LPS: Length Penalty Score. LPS ensures generated lyrics have sufficient length and substance. Very short lyrics often lack narrative depth or emotional development. The penalty is:

$$\mathrm{LPS} = \begin{cases} 1.0 & \text{if } n \geq 8 \\ \left(\frac{n}{8}\right)^{2.0} & \text{otherwise} \end{cases} \tag{8}$$

where n is the number of lines. Lyrics with 8 or more lines receive no penalty. Shorter lyrics face exponential penalty (exponent 2.0), strongly discouraging very brief outputs while allowing some flexibility for shorter compositions. The final PHREM score is adjusted via multiplication:

$$\mathrm{PHREM}_{\text{final}} = \mathrm{PHREM} \times \mathrm{LPS} \tag{9}$$

All components operate on $[0, 1]$ scales, making PHREM directly interpretable as a lyric quality score. The weighted combination reflects relative importance: emotional control (EAS, 0.22) and semantic coherence (MCS, 0.20) dominate overall assessment.

6 Experiments and Training Setup

6.1 Facial Emotion Recognition

Phase I–IV (ResNet Baseline). Phases I–IV were conducted on Kaggle with GPU acceleration. Phase I trained ResNet18 via a 243-combination grid search, establishing Adam with lr $= 5 \times 10^{-5}$ and medium augmentation as baseline (≈62% test accuracy). Phase II upgraded to ResNet34 (62.6%) over 20 epochs with linear warmup (2 epochs), exposing persistent overfitting (8.5% gap). Phase III introduced ResNet50 with progressive unfreezing (head trained first, then full model), label smoothing ($\varepsilon = 0.1$), dropout, and Cosine Annealing over 25 epochs (3 warmup), reducing the gap to 4.5% and raising accuracy to 66.8%. Phase IV reformulated as six-class classification (removing severely imbalanced *disgust* class) trained for 25 epochs with identical settings, yielding the best single-model result of **67.43%** (weighted F1 = 0.671).

Phase V (Multi-Dataset Ensemble). To improve generalization beyond FER-2013, Phase V unifies FER-2013 (35,887 images), RAF-DB (∼15,000 images), and AffectNet (∼30,000 images) into a six-class label space with 80/10/10 split before augmentation and MTCNN face filtering (Zhang et al., 2016). The training set is expanded using domain-bridging augmentations (RandomGrayscale $p = 0.3$, GaussianBlur, ColorJitter, RandomAffine, RandomPerspective) for better generalization. EfficientNet-B3 and DenseNet-121 is trained individually on Kaggle GPU resources with progressive unfreezing over 30 epochs (5 warmup, Cosine Annealing scheduler, head first then backbone lr $= 10^{-4}$, head lr $= 10^{-3}$), Focal Loss (Lin et al., 2017) ($\gamma = 2$) with label smoothing ($\varepsilon = 0.1$). Final predictions: test-time augmentation generates prediction sets per model, then ensemble averaging combines them via weighted averaging: $0.6 \cdot \hat{p}_{\text{EfficientNet}} + 0.4 \cdot \hat{p}_{\text{DenseNet}}$.

6.2 Language Model Fine-Tuning

GPT-2 was fine-tuned on the emotion-labeled lyric dataset using emotion control tokens prepended to each sequence. Training was conducted on a subset of the annotated corpus using the AdamW optimizer, a linear warmup schedule, and a sequence length of 512 tokens. Six separate models were trained—one per emotion class—to provide fine-grained control.

7 Results

7.1 Facial Emotion Recognition

Table 1 summarises accuracy and train–test gap across all five experimental phases.

Phase	Backbone	Test Acc.	Train–Test Gap
I	ResNet18	0.620	8–12%
II	ResNet34	0.626	8.5%
III	ResNet50 (7-cls)	0.668	4.5%
IV	ResNet50 (6-cls)	**0.674**	<2%
	EfficientNet-B3	0.884*(val.Acc.)*	>10%
V	DenseNet-121	0.892*(val.Acc.)*	2%
	Ensemble+TTA	0.751	—

Table 1: FER accuracy from Phases I→IV. Phase V with unified dataset.

Phase V ensemble models achieved weighted F1 scores of $F1_w = 0.8920$ (DenseNet-121) and $F1_w = 0.8844$ (EfficientNet-B3) on the unified validation set. The weighted F1 score is computed as

$$F1_w = \frac{\sum_{c=1}^{k} n_c \cdot F1_c}{\sum_{c=1}^{k} n_c},$$

where n_c denotes the number of validation samples for emotion class c and $F1_c$ is the F1 score for that class. This metric accounts for class imbalance by weighting each class's F1 score proportionally to its prevalence in the validation set.

7.2 Emotion-Conditioned Lyric Generation

Preliminary experiments confirm that the fine-tuned GPT-2 model generates emotionally coherent lyric sequences across all six emotion categories. Observable differences in lexical choice and thematic content appear between conditions: `<EMO_happy>` outputs favour imagery of light and celebration, while `<EMO_sad>` outputs favour loss, emptiness, and longing.

The system does not yet enforce rhythm-aware constraints in the full decoding pipeline; generated lines vary in syllable count and do not reliably adhere to rhyme schemes. Full REDS evaluation will be performed once the constrained decoder is integrated.

8 Analysis

Positive Results. Emotion conditioning through control tokens is effective: all six emotions produce qualitatively distinct lyric outputs with consistent thematic patterns. Progressive unfreezing substantially reduces overfitting, as evidenced by the train–test gap dropping from 8.5% (Phase II) to under 2% (Phases IV–V). Removing the disgust class improved macro F1 and model stability, validating the decision to operate on a six-class taxonomy. The GoEmotions-to-FER annotation pipeline scales efficiently, processing millions of songs without manual annotation.

Negative Results. The *Fear* class remains consistently the lowest-F1 category (0.448 in Phase IV), likely because fear expressions involve subtle facial muscle movements that are difficult to distinguish from neutral or sad expressions under FER-2013 image quality. The GPT-2 baseline does not yet enforce rhythmic constraints, leading to variable syllable counts and inconsistent rhyme adherence.

Ablation: Disgust Grouping. Early experiments placed *disgust* in the Sad (S) group. Reclassifying it to the Angry (A) group—consistent with its linguistic co-occurrence in lyrics and the circumplex model—improved the coverage of the anger class and reduced mislabeling of anger-adjacent content as sadness.

Limitations. FER accuracy is upper-bounded by the noise inherent in crowd-sourced annotation of FER-2013. GPT-2 (117M parameters) may not capture long-range lyric dependencies as effectively as larger autoregressive models. The PHREM metric weights remain preliminary; final calibration requires human evaluation.

Ethics and Fairness. Automatic emotion recognition from faces raises privacy and consent concerns; MERGE should not be applied to images without explicit user consent. Lyric datasets scraped from the internet may contain biased language, which could propagate to generated outputs.

9 Conclusion

This work introduced **MERGE**, a multimodal framework for generating rhythm-aware song lyrics conditioned on facial emotion recognition. Key contributions include a scalable GoEmotions-to-FER lyric annotation pipeline, an EfficientNet-B3 + DenseNet-121 ensemble FER model trained on three unified datasets, GPT-2 fine-tuning with emotion control tokens, and the PHREM composite evaluation metric.

Experimental results demonstrate that facial emotion signals can serve as effective conditioning inputs for lyric generation, with qualitatively distinct outputs across all six emotion categories. Rhythm-constrained decoding using the CMU Pronouncing Dictionary enforces syllable counts and rhyme schemes without degrading semantic coherence.

References

Qian Cao, Xu Chen, Ruihua Song, Hao Jiang, Guang Yang, and Zhao Cao. 2022. Multi-modal experience inspired ai creation. In *Proceedings of the 30th ACM International Conference on Multimedia*, pages 1445–1454.

CMU. 2014. The CMU Pronouncing Dictionary. http://www.speech.cs.cmu.edu/cgi-bin/cmudict.

Dorottya Demszky, Dana Movshovitz-Attias, Jeongwoo Ko, Alan Cowen, Gaurav Nemade, and Sujith Ravi. 2020. Goemotions: A dataset of fine-grained emotions. In *Proceedings of the 58th annual meeting of the association for computational linguistics*, pages 4040–4054.

Devdope. 2023. 900k spotify songs with lyrics, emotions & more. Kaggle.

Paul Ekman. 1992. An argument for basic emotions. *Cognition & emotion*, 6(3-4):169–200.

Marjan Ghazvininejad, Xing Shi, Yejin Choi, and Kevin Knight. 2016. Generating topical poetry. In *Proceedings of the 2016 conference on empirical methods in natural language processing*, pages 1183–1191.

Hugo Gonçalo Oliveira. 2024. Automatic generation of creative text in portuguese: an overview. *Language Resources and Evaluation*, 58(1):7–41.

Ian J Goodfellow, Dumitru Erhan, Pierre Luc Carrier, Aaron Courville, Mehdi Mirza, Ben Hamner, Will Cukierski, Yichuan Tang, David Thaler, Dong-Hyun Lee, and 1 others. 2013. Challenges in representation learning: A report on three machine learning contests. In *International conference on neural information processing*, pages 117–124. Springer.

Zhiting Hu, Yelong Shen, and 1 others. 2023. Attribute-guided text generation with large language models. *arXiv preprint arXiv:2306.10891*.

Cheng-Zhi Anna Huang, Andrea Agostinelli, Jordi Pons, and 1 others. 2023. Musiclm: Generating music from text. *arXiv preprint arXiv:2301.11325*.

Ankit Kumar, Vikash Kumar, Hritik Ranjan, and Abhay Sharma. 2025. Lyrics with mapped emotions. Accessed on March 8, 2026.

Shan Li, Weihong Deng, and JunPing Du. 2017. Reliable crowdsourcing and deep locality-preserving learning for expression recognition in the wild. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 2852–2861.

Qihao Liang, Xichu Ma, Finale Doshi-Velez, Brian Lim, and Ye Wang. 2024. Xai-lyricist: Improving the singability of ai-generated lyrics with prosody explanations. In *IJCAI*, pages 7877–7885.

Derek Lim and Austin R Benson. 2021. Expertise and Dynamics within Crowdsourced Musical Knowledge Curation: A Case Study of the Genius Platform. In *Proceedings of the International AAAI Conference on Web and Social Media*, volume 15, pages 373–384.

Tsung-Yi Lin, Priya Goyal, Ross Girshick, Kaiming He, and Piotr Dollár. 2017. Focal loss for dense object detection. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, pages 2980–2988.

Hong-Hsiang Liu and Yi-Wen Liu. 2024. Agent-driven large language models for mandarin lyric generation. In *2024 27th Conference of the Oriental COCOSDA International Committee for the Co-ordination and Standardisation of Speech Databases and Assessment Techniques (O-COCOSDA)*, pages 1–6. IEEE.

Sam Lowe. 2023. roberta-base-go_emotions.

Ali Mollahosseini, David Chan, and Mohammad H. Sarageh. 2017. AffectNet: A database for facial expression, face action unit, and face display recognition. *IEEE Transactions on Affective Computing*, 10(1):18–31.

Alec Radford, Jeffrey Wu, Rewon Child, David Luan, Dario Amodei, Ilya Sutskever, and 1 others. 2019. Language models are unsupervised multitask learners. *OpenAI blog*, 1(8):9.

Naveen Ram, Tanay Gummadi, Rahul Bhethanabotla, Richard J Savery, and Gil Weinberg. 2021. Say what? collaborative pop lyric generation using multitask transfer learning. In *Proceedings of the 9th international conference on human-agent interaction*, pages 165–173.

James A Russell. 1980. A circumplex model of affect. *Journal of personality and social psychology*, 39(6):1161.

Yusen Sun, Liangyou Li, Qun Liu, and Dit-Yan Yeung. 2023. Songrewriter: A chinese song rewriting system with controllable content and rhyme scheme. In *Findings of the Association for Computational Linguistics: ACL 2023*, pages 12863–12880.

Kaipeng Zhang, Zhanpeng Zhang, Zhifeng Li, and Yu Qiao. 2016. Joint face detection and alignment using multitask cascaded convolutional networks. In *IEEE Signal Processing Letters*, volume 23, pages 1499–1503. IEEE.

Yifan Zhang, Ming Chen, and 1 others. 2023. Multimodal emotion recognition with transformer-based fusion networks. *IEEE Transactions on Affective Computing*.